# Fact-Checkers Navigating Generative AI: Practices, Boundaries, and Design Implications

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The rapid and large-scale spread of misinformation poses a serious challenge for fact-checkers, whose work remains difficult to scale and is often resource-intensive. In response, fact-checking organizations are increasingly adopting generative AI to support verification work. While these tools may improve efficiency and expand capacity, their use also raises concerns about fairness, accountability, transparency, and factuality. Here, we provide an empirically grounded account of how professional fact-checkers integrate generative AI into high-stakes verification work, showing that adoption is heterogeneous, constrained by professional norms and structural conditions, and shaped by conflicts with fairness, accountability, and transparency. We present findings from semi-structured in-depth interviews with 29 professional fact-checkers from 28 organizations operating in 41 countries. Our analysis shows substantial variation in how fact-checkers engage with generative AI, ranging from interaction to the development of in-house systems and public-facing LLM-powered fact-checking chatbots. Across these practices, participants position LLMs as assistive tools embedded in human-led workflows, drawing clear boundaries around automation to preserve core fact-checking principles. At the same time, they report persistent challenges, including hallucinations, uncertainty introduced by probabilistic model outputs, difficulties in source traceability, and uneven performance across languages. Adoption is further shaped by structural constraints, including access restrictions, costs, and organizational capacity. Participants also describe shifts in the verification ecosystem as audiences increasingly rely on AI chatbots to fact-check claims directly. Grounded in fact-checkers' perspectives, this study contributes an empirically informed account of human–AI interaction in professional fact-checking and derives user-centered design considerations.

CCS Concepts: • **Human-centered computing → Empirical studies in HCI**; • **Social and professional topics →** *Socio-technical systems*; • **Computing methodologies → Artificial intelligence**.

Additional Key Words and Phrases: Generative AI, Large Language Models, Fact-checking, Fact-checkers, Human–AI Interaction, Misinformation

**ACM Reference Format:**
Basak Bozkurt, Mohsen Mosleh, and Helen Margetts. 2026. Fact-Checkers Navigating Generative AI: Practices, Boundaries, and Design Implications. In *The 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26), June 25–28, 2026, Montreal, QC, Canada.* ACM, New York, NY, USA, 24 pages. https://doi.org/10.1145/3805689.3806474

## 1 Introduction

The rapid and large-scale spread of misinformation poses a significant challenge that can undermine public discourse, erode trust, and distort democratic decision-making [5, 62]. Fact-checking has emerged as a response to address these challenges. However, the scale and speed of misinformation often exceeds the capacity of

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*FAccT '26, Montreal, QC, Canada*

ACM ISBN 979-8-4007-2596-8/2026/06
https://doi.org/10.1145/3805689.3806474

fact-checkers, whose work remains largely manual [6, 44]. This challenge has been further exacerbated by the rise of large language models (LLMs), as they can misinform the public on a broad scale, due to misuse, tendencies to hallucinate, their reliance on outdated data, or a lack of domain expertise [8, 43, 67, 71, 90]. Moreover, the fluent and persuasive style of LLM-generated content, often accompanied by a confident tone, can make false and misleading information more convincing to users [8], further increasing the burden on fact-checkers.

Despite these challenges, LLMs can help combat misinformation. Trained on vast amounts of data across diverse topics and equipped with rapid information retrieval capabilities [16], LLMs can potentially be good at fact-checking [23]. Previous research suggests that LLMs can assist fact-checkers at various stages of their work, including detecting check-worthy claims, predicting potential harm, identifying previously fact-checked claims, crafting explanations, detecting stances, and offering multilingual support, summarization, and transcription [8, 23, 25].

Much of the existing research has examined LLMs as autonomous agents through benchmark performance and general capabilities. Less attention has been paid to their role as user-facing tools in human–AI interaction, a gap that gains importance as LLMs are increasingly adopted across both everyday and professional domains [88]. This gap is particularly salient in professional fact-checking, where LLMs are incorporated into workflows that demand high levels of accuracy, fairness, accountability, and transparency.

Although generative AI is now increasingly visible in fact-checking practice, empirical research on how professional fact-checkers use LLMs remains limited [29]. To date, only a small body of work has focused on the implications of generative AI for professional fact-checking [27, 91, 95]. However, these studies reflect earlier characterizations of adoption and experimentation that no longer fully capture current practice. Building on this emerging body of research, this study provides an updated, practice-oriented account of how LLMs are currently being integrated into fact-checking work across several organizations. Our study examines how professional values, norms, and practical constraints shape the integration of generative AI into fact-checking practice. This paper asks the following research questions:

(1) **RQ1:** How do fact-checkers use generative AI, and what benefits, challenges, and needs do they identify?
(2) **RQ2:** How do fact-checkers perceive the impact of generative AI on fact-checking practices?
(3) **RQ3:** What factors shape the ways professional fact-checkers use generative AI?

To address these research questions, we conducted in-depth interviews with 29 participants from 28 fact-checking organizations operating in 41 countries across five continents. Drawing on this diversity of contexts, this study makes five contributions: (1) it provides an empirical account of how professional fact-checkers incorporate LLMs into verification workflows, considering differences in organizational capacity, resources, and regional contexts; (2) it examines how the use of generative AI aligns (or fails to align) with commitments to fairness, accountability, and transparency in professional verification work; (3) it provides one of the first detailed examinations of LLM-based chatbots developed by fact-checking organizations as public-facing tools for distributing fact-checks; (4) it extends prior research that has largely focused on the Global North [22] by expanding the scope to include perspectives from the Global South; and (5) it translates findings into design implications. Together, this study contributes to scholarship on fact-checking and human–AI interaction and informs the development of AI-assisted verification practices that better align with fact-checkers' workflows and professional norms.

## 2 Related Work

### 2.1 The Work Practices of Fact-checkers

Fact-checking originated in early journalism, and it mainly involved verifying claims in news articles before publication [19]. In recent decades, it expanded to include verifying claims in already published information [39], becoming "retroactive gatekeeping" [86]. This has led to the creation of dedicated fact-checking organizations and the formation of fact-checking units within existing media companies. The first of these, FactCheck.org, was established in the U.S. in 2003 [39]. Similar organizations have since emerged worldwide. From around 2016 onward, the number of organizations increased significantly, driven by concerns about disinformation, particularly following the 2016 U.S. presidential election and the Brexit referendum [40, 41, 87]. In early January 2026, based on the Duke's Reporters' Lab database, 450 fact-checking organizations were active worldwide [31], with about 35% of them being signatories to the International Fact-Checking Network (IFCN) [53]. Collectively, these fact-checkers contribute to global anti-misinformation efforts. This scale is evidenced by ClaimReview [37], a database containing more than 200,000 verified claims from fact-checkers around the world [82].

Fact-checking has developed into a recognized profession with its own set of norms and practices, alongside the emergence of regional [2, 32] and global organizations that formalize these standards [51]. At the global level, the IFCN plays a central role as a well-established body that defines a widely adopted code of principles, including fairness, transparency, methodological rigor, and an open corrections policy in fact-checking. Fairness refers to non-partisanship and the consistent application of the same standards and procedures across all fact-checks, regardless of political affiliation or ideological position. Fact-checking organizations are expected to be transparent, disclosing sources in detail to enable independent verification and replication of their findings, as well as to openly declare funding sources. When errors occur, organizations are expected to publish corrections [52].

Methodological rigor requires organizations to clearly articulate how claims are selected, researched, written, published, and corrected [52]. In practice, fact-checkers typically follow a structured methodology divided into five stages [39, 73]: (1) selecting widely circulated claims or those with a higher potential to mislead the public; (2) contacting the individual who made the claim to clarify intent and gather context; (3) tracing and analyzing sources for origin and credibility, as well as examining how the claim may have been modified or misinterpreted; (4) consulting experts for data interpretation; and (5) writing up the findings in a transparent manner, supported by appropriate evidence. These steps are more often used for scrutinizing claims from politicians, which was the original aim of fact-checking.

Since 2016, there has been a shift toward debunking misinformation directly on social media platforms. As part of this shift, fact-checking organizations have collaborated with platforms through third-party partnership programs [11, 54, 74] where they flag false content and are compensated for this work. In recent years, however, platforms have moved away from professional fact-checking and toward community-based verification models. For example, Meta discontinued its third-party fact-checking program in the U.S. in 2025 [58], and this move has also raised questions about the future of platform–fact-checker partnerships in other regions [42].

## 2.2 Fact-checking as a Socio-technical Practice

Fact-checking is a socio-technical practice [92, 93] that requires human judgment, while also relying on technology to collect, analyze, and verify information in order to improve efficiency and scale [9, 57, 73]. As a result, fact-checking work includes the interaction between human expertise and technical systems. Previous research has examined the human dimension of these interactions, including the role of fact-checkers, editors, researchers, social media managers, and audiences [57]. Beyond these roles, the human infrastructure of fact-checking also includes technical personnel who design, develop, and maintain AI systems used in verification work [22].

Fact-checkers commonly use off-the-shelf tools, such as spreadsheets, search tools, and social media monitoring tools, alongside platform-provided tools [18, 34, 69], including Meta's AI system for flagging potentially false claims [72], and image and video verification tools, such as Google reverse image search, TinEye, and InVID [57]. These tools are particularly beneficial for fact-checking organizations, many of which lack the in-house technical expertise needed to develop customized solutions [73]. Despite these benefits, reliance on external and fast-changing tools introduces several challenges. Fact-checkers often rely on multiple tools for a single task, complicating workflows [73]. These tools are dynamic; their capabilities are subject to change, and change does not always equate to improvement [69]. Fact-checkers face disruptions resulting from modifications or the discontinuation of these tools [38, 57, 69]. Fact-checkers, particularly in the Global South, often face further limitations, such as poor or missing language support and scarce resources for building tools that work in their regional languages [46, 57]. Many lack the expertise to customize tools to their requirements [34, 73].

In parallel, there has been growing academic interest in the design and development of automated fact-checking systems. However, the practical applicability of these systems remains limited. They often reflect technical possibilities rather than professional needs [28, 75, 84]. By framing fact-checking as a purely technological problem, such approaches risk overlooking the socially situated and collaborative nature of fact-checking practice [57]. Fact-checkers remain skeptical about fully automating the entire process [57, 79], as many core fact-checking tasks, such as interpreting complex claims and contacting experts, are fundamentally human-centered processes. Fact-checkers, therefore, position themselves as humans in the loop and remain responsible for determining final verdicts [79].

A growing body of qualitative research has examined fact-checkers' perceptions of AI and automation more broadly [34, 49, 57, 61, 73, 79, 83]. Empirical research on AI use in fact-checking has highlighted that organizations often engage with AI systems primarily as end users and have limited opportunities to experiment with or develop their own AI systems [61]. Where such efforts exist, the integration of automation tools into fact-checking workflows is reported to be at an early stage and limited in scope [73]. However, how fact-checking organizations engage with AI continues to evolve [22], suggesting that earlier characterizations of limited and early-stage adoption may no longer fully capture current practice. One study mapping the opportunities and challenges associated with generative AI use was conducted in late 2023 and early 2024, capturing a period when fact-checking organizations were only beginning to experiment with LLMs. This work provides early insights into emerging practices and perceptions through an organizational adaptation lens [95]. Subsequent work by the same researchers further shows that fact-checking organizations are making strategic choices between open and proprietary models based on trade-offs, including cost, performance, and safety [96]. Another early study focused on the perceptions of fact-checkers regarding LLMs in Spanish fact-checking organizations. Related work on explainable automated fact-checking has also provided evidence of an ongoing transition toward the use of LLMs in fact-checking [91].

These relatively early studies have been valuable in highlighting initial adoption patterns and the potential of LLMs, often framing their use from an end user perspective. However, the technology continues to evolve rapidly, underscoring the need for updated analyses that capture how emerging practices take shape. Moreover, fact-checking organizations are not homogeneous in how they engage with LLMs. Beyond relying on off-the-shelf tools, some organizations actively develop, customize, and maintain their own systems, integrating LLMs into workflows through prompting, in-house tools, agent-based systems, and public-facing chatbots (e.g., [7, 24, 26]). Despite their increasing prevalence, these practices remain underexplored in existing research. This study aims to address these gaps.

### 2.3 Generative AI in Fact-checking

The fact-checking pipeline consists of three stages: claim detection, evidence gathering, and claim verification, which includes verdict prediction and justification production [3, 44]. These stages are not only technical steps, but also stages where epistemic judgment and accountability are exercised: deciding which claims merit attention, what sources count as evidence, and how to justify conclusions. In this sense, fact-checking systems raise fairness, transparency, and accountability concerns, not only through model performance but also through how they shape access to evidence, allocate attention, and distribute error and responsibility across actors.

*Claim detection* involves identifying check-worthy claims, a binary classification task assessing criteria such as verifiability, harmfulness, or social impact [4, 48]. This stage also includes *claim matching*, which retrieves previously fact-checked claims, typically via ranking based on semantic and lexical similarities [85]. *Evidence retrieval* includes gathering relevant information to evaluate a claim, while *claim verification* entails verdict prediction and justification production [44]. LLMs can augment each stage of this pipeline. In the *claim detection* stage, LLMs can assess the check-worthiness of claims [47] and identify previously fact-checked claims through zero-shot, few-shot, and chain-of-thought prompting [25]. In the *evidence retrieval*, LLMs can assist with summarization and information retrieval stage, as well as with generating explanations [23]. Factuality remains a persistent challenge when using LLMs for fact-checking. LLMs often produce answers without clear or verifiable sources and can generate fluent but incorrect information. Their confident and fluent style may lead users to trust these outputs too easily, and limitations in up-to-date knowledge further reduce their accuracy [8, 89]. To improve the factuality of LLMs, researchers have explored various approaches, such as Retrieval-Augmented Generation (RAG) [59, 63] and in-context learning [99]. However, how frequently these challenges occur and how effectively these approaches are adopted in fact-checking workflows remains unclear. Our study examines how fact-checkers use LLMs across these workflow steps, how factuality challenges surface in real-world fact-checking, and how fact-checkers respond to them in everyday verification work.

### 2.4 Human–AI Interaction in the LLM Era

By demonstrating unprecedented abilities, such as generating human-like text and functioning as conversational agents, LLMs challenge earlier expectations of machine behavior and mark a notable shift within the long-established field of human–AI interaction [98]. Previous work has conceptualized human–AI interaction as a spectrum of collaboration rather than full automation, in which AI systems vary in autonomy while humans retain oversight, control, and responsibility. One study described AI roles as ranging from tools and servants that carry out well-defined, low-level tasks (e.g., information processing or sorting) to assistants and mediators that exhibit greater autonomy. In such a setting, AI systems may suggest content or support decision-making, while humans retain ultimate control and responsibility [60]. Focusing specifically on LLMs, recent work categorizes Human–LLM interaction into four patterns [65]: (1) processing tools, in which LLMs execute predefined tasks to support human decision-making; (2) analysis assistants, where LLMs help analyze user input while humans make final decisions; (3) creative companions, in which humans and LLMs collaboratively generate ideas and make decisions; and (4) processing agents, where humans provide initial goals and LLMs autonomously carry out tasks. In addition to these outcome-oriented frameworks, there are also classifications of human–LLM interaction according to interface design, such as standard prompting, structured user interfaces with tools such as forms and visualizations, context-based interactions using explicit specifications or role-play and example-based prompting, and agent facilitator modes in which the LLM facilitates tasks within a team [35]. Although these frameworks offer valuable ways of conceptualizing human–LLM interaction, little is known about how such interaction patterns manifest in high-stakes settings such as professional fact-checking. Our study extends this line of work by examining how fact-checkers engage with LLMs in practice.

## 3 Methods

### 3.1 Participant Recruitment

We conducted 28 semi-structured interviews with 29 professional fact-checkers from IFCN-member fact-checking organizations, between May and September 2025. One interview involved two participants from the same organization. Participants came from fact-checking organizations operating in 41 countries across five continents (see Figure 1). Among the participants, 17 identified as women and 12 as men, ranging in age from 18 to over 55, with most (13) between 25 and 34 years old. Participants represented a range of career stages, though most were early- to mid-career practitioners. The majority held bachelor's or master's degrees and worked in a range of roles, including research, editorial, and management positions. Participant demographics are provided in Table 1 in Appendix A.1. Details of the recruitment strategy and materials are provided in Appendix A.2.

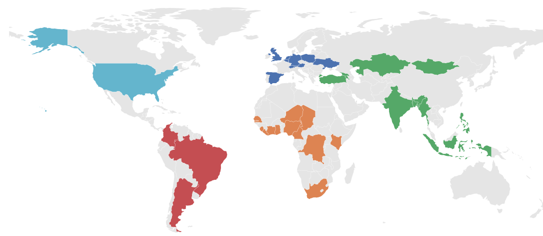

Fig. 1. The study included 29 participants from 28 fact-checking organizations operating across 41 countries and five continents, with several organizations operating in more than one country.

### 3.2 Data Collection

Eligibility criteria required participants to be over 18, to use generative AI in their work, and to be employed at an IFCN signatory organization. Following recruitment, eligible participants who completed the registration form were contacted (see Appendix A.3), and interviews were scheduled. Participants were also asked to complete a separate demographic questionnaire (see Appendix A.4) before the interview. The interviews followed a semi-structured format, guided by a set of interview questions (see Appendix A.5). All interviews were conducted remotely via Microsoft Teams and typically lasted around one hour. With participants' oral consent, all interviews were recorded and automatically transcribed using Microsoft Teams' built-in transcription function. As a token of appreciation, participants were offered a £35 online gift card. Three participants chose to donate their compensation to the IFCN, and three opted to donate it to a fact-checking organization of their choice.

### 3.3 Data Analysis

We conducted a reflexive thematic analysis following the six-phase guide outlined by Braun and Clarke [15]. Phase 1 involved reading the interview transcripts, cross-checking them against the original recordings for accuracy, taking notes, and recording initial ideas for coding. In Phase 2, initial codes were generated across all transcripts using NVivo 14. We employed both semantic coding, which captured explicit meanings based on what participants said, and latent coding, which involved interpretive analysis of underlying assumptions and meanings beyond the descriptive content [17]. We followed an inductive approach, in which codes were developed during analysis [12]. Phase 3 focused on generating themes by grouping similar codes into patterns of shared meaning [13]. Phase 4 consisted of a two-step process for developing and refining themes: first, checking coherence within themes, and then ensuring that themes accurately represented the entire data set and adjusting them as needed [15]. This step was finalized once further refinements no longer added meaningful value. Phase 5 included the final refinements, where the main message of each theme was clarified (see Appendix B for themes and their scope). Finally, Phase 6 involved writing up the analysis. Interview quotes are presented as near-verbatim quotes, with minor edits for clarity and readability (e.g., the removal of repetitions and filler words) while preserving participants' original meaning.

## 4 Findings: Professional Fact-Checkers' Uses, Needs, and Perceptions of LLMs

Our analysis identified five themes: (1) human-led fact-checking with LLM assistance, describing how LLMs are selectively used across different stages of fact-checking while verification remains human-led; (2) fact-checking principles in conflict with generative AI, including concerns related to fairness, accountability, and transparency; (3) factuality challenges in LLM-assisted fact-checking; (4) conditions of adoption: access, cost, and organizational capacity, which shape how LLMs are taken up in practice; and (5) shifts in the verification ecosystem. These themes are discussed below.

### 4.1 Human-led Fact-checking with LLM Assistance

Participants described workflows that broadly matched established accounts of professional fact-checking, including claim identification, evidence gathering, verification, and publication. Generative AI tools were described as being incorporated into specific stages of the process. In claim detection, LLMs are used to monitor large volumes of data, transcribe content, identify check-worthy claims, and highlight signals related to potential harm. During evidence gathering, participants reported using LLMs to support searching, transcription, and summarization, and to assist with data analysis. Several participants also highlighted their time-saving benefits during preliminary research. As one participant explained:

> Whenever we are doing research via Google or other tools, we usually need more time, because when you're Googling something, you still need to check every website and every document, and it takes a lot of time. But (Chat)GPT or kind of (similar chatbots) summarize this information for you. (P01)

LLMs are also used in downstream stages of distribution to speed up writing, support content generation for social media, and create visual content. Participants mentioned that AI tools were also used to brainstorm. One participant described using LLMs in training to help interns recall relevant questions or investigative steps (P07).

Across these stages, participants emphasized that LLMs are mainly used to save time and support productivity. At the same time, participants articulated clear boundaries around generative AI use within fact-checking workflows. They rejected the idea of fully automating fact-checking; instead, they framed generative AI as a tool to support research and productivity, thereby allowing fact-checkers to focus on interpretation and judgment. LLMs are explicitly excluded from writing full fact-checking articles or serving as evidential sources, and their use in the verification stage is described as highly limited or, in some cases, absent. Several participants distinguished between what P24 referred to as "fast checks," in which LLMs are used to quickly match claims against available or previously verified sources, and full "fact-checks," which require human-led verification and, where necessary, escalation to expert judgment, particularly in cases involving complex claims or novel issues. Engagement with LLMs was therefore characterized by conditional adoption rather than full acceptance or rejection.

### 4.2 Fact-checking Principles in Conflict with Generative AI

*4.2.1 Fairness.* Participants reported uneven generative AI performance across different languages and regions, which was particularly evident in low-resource languages. In these environments, generative AI was described as less reliable when verification depended on local knowledge, cultural references, or language-specific nuance. Automated transcription was frequently reported to struggle with capturing everyday speech. These limitations were especially pronounced in speech-to-text and code-switching scenarios, where transcription errors not only distorted meaning but, in some cases, introduced new misinformation into verification workflows (P15). These challenges were most acute in many African countries, where radio and audio content often serve as primary sources of claims. Similar difficulties were reported for languages with non-standardized written forms, such as Cantonese, where spoken meanings are frequently lost or altered during transcription (P21).

Participants (P09, P14, P15, P19) further noted that even when speakers use English, "the accuracy is low because of the way we speak here. The way people in (their country) speak English is different (...) and we're always mixing our local language into how we speak" (P09). In response to these shortcomings, one organization developed its own transcription tools trained on local accents, which led to performance improvements.

Beyond transcription, participants also observed that identical queries posed in different languages could produce varying results, sometimes drawing on biased or lower-quality sources in one language but not another (P13, P28). From this perspective, language functioned not just as a technical input, but as an epistemic filter that shaped what counted as evidence. Participants interpreted these asymmetries as reflecting broader imbalances in model training data and development priorities.

Overall, uneven AI performance across languages and regions raises important fairness concerns because it shapes which claims can be effectively verified and which forms of evidence become visible and credible. These findings highlight the continued importance of human expertise and judgment in fact-checking, particularly in settings where linguistic nuance and local knowledge are essential.

*4.2.2 Accountability.* Participants emphasized the importance of keeping humans in the loop in AI-assisted fact-checking workflows, which we interpret through the lens of accountability. Whether AI tools were used for research, transcription, or content creation, participants underscored that fact-checkers are fully accountable for the accuracy and implications of the final output. Several rejected the idea of treating AI systems as independent decision-makers, arguing that responsibility cannot be delegated to opaque technologies. As P02 said: "We have rules, for example, always (have a) human being in the loop, (a) human being at the beginning, (and a) human being at the end. You are responsible for your story. Get your facts right." This illustrates how participants understand verification as a responsibility that lies with the fact-checker.

*4.2.3 Transparency.* Participants described professional fact-checking norms as grounded in norms of transparency, including source disclosure, explainable reasoning, and editorial responsibility. While acknowledging that human fact-checkers are not free from bias, they emphasized that these practices make underlying assumptions and limitations explicit, allowing them to be scrutinized, challenged, and corrected. In contrast, participants noted that LLMs are opaque, as it is difficult to trace the sources of the information they provide, understand the reasoning behind their outputs, or obtain consistent results due to their probabilistic nature. These limitations were perceived as being in conflict with the principle of transparency.

During interviews, discussions of transparency also focused on practices of AI disclosure. On the one hand, participants noted that distinguishing what counts as AI use has become increasingly difficult, a challenge compounded by the absence of clear guidelines. Participants noted that functionalities such as image verification, transcription, visualization, and content production now increasingly incorporate generative AI, often without being explicitly recognized as AI by users. As P15 observed, existing AI policies leave "gray areas" around disclosure for everyday uses such as headline writing, making it unclear how transparency should be applied in practice.

Across interviews, we observed the absence of clear guidelines for the use of generative AI in fact-checking. As a result, organizations relied on self-regulation, with decisions around disclosure often left to the interpretation of individual fact-checkers. As one participant explained, organizational policies were being developed in tandem with experimentation:

> We're cautious about adoption moving ahead of policy, so our policies are being developed alongside experimenting with these tools and understanding what they actually look like in our workflows. We're trying to anticipate what needs to be considered in terms of ethical and responsible use, including accuracy and transparency with audiences. But we're still figuring out what that looks like in practice. AI is such a blanket term, and it's being integrated into tools we've used for years, so where does our responsibility for transparency begin and end? How much detail do we go into when we say we used AI? (P19)

This account highlights how the lack of formal guidance shifts responsibility for transparency onto individual practitioners, while also illustrating the increasingly blurred boundaries around where to draw the line as more tools incorporate generative AI.

### 4.3 Factuality Challenges in LLM-assisted Fact-checking

Participants identified hallucination as the most serious factuality challenge (P01, P05, P07, P13-14, P16, P18-19, P23-24), especially during research. Like P01, many others described similar experiences: "We usually ask it (ChatGPT) to provide sources, because we have had cases where it just made up information. For example, when we were looking for research articles about a certain topic, it just made up the research articles and the authors." Participants shared a range of strategies to manage hallucinations, including manual verification, cross-checking outputs, and querying the information with multiple LLMs. Some organizations also developed systems based on RAG, combined with prompt-level guardrails that require models to acknowledge uncertainty or return no answer when supporting evidence is insufficient (P05, P06, P08, P09, P11, P17, P26). One participant explained how adopting a RAG-based design directly reduced hallucination risks, while also improving the user interaction:

> We came up with a beta version (of the chatbot) that included LLMs in the pipeline, and RAG kind of solved the biggest concerns we had, especially hallucinations and how fast we could update the chatbot. What we did was connect the LLM directly to our database of fact-checks. Every time a user asks a question, we search our database for relevant context and then feed that context to the model, asking it to use only the information we provide. In terms of user experience, this changed things a lot compared to the first version. Instead of relying on pre-made rules, users can now talk naturally to the chatbot and get answers that directly address the questions they ask. (P26)

By constraining model outputs to verified sources, the RAG-based design helped address hallucination risks and increase confidence in the system's responses. This design also improved audience experience: the same participant reported an increase in positive user feedback from 40% for the rule-based WhatsApp chatbot to 75% for the RAG-based system.

Participants discussed accuracy as a central concern in the use of LLMs for fact-checking, while noting that acceptable accuracy levels depend on audience and user context. As one participant explained, their users are "expert fact-checkers," and if such systems were made available to the public, "we'd (they'd) be adopting a different process." In professional settings, imperfect accuracy was described as manageable, provided that systems were designed to make it easy for fact-checkers to question or dismiss outputs: "I don't have to have 100% accuracy… we have a 90% accuracy, but we make it easy for people to dismiss it" (P25). This suggests that acceptable accuracy thresholds are context-dependent, varying by user expertise and use case.

### 4.4 Conditions of Adoption: Access, Cost, and Organizational Capacity

Beyond how generative AI is used in practice, its adoption is also shaped by a set of structural conditions: access to tools, costs of deployment, and the organizational capacity needed to build and sustain them. Together, these conditions produce uneven adoption across fact-checking organizations.

*4.4.1 Access.* Several participants saw generative AI as necessary for scaling fact-checking efforts and ensuring access to adequate, up-to-date tools. As P25 put it: "fact checkers deserve access to the best technology." This is because they are the "first responders" dealing with "some of the worst information on the Internet," whose outputs are reused for model training and content moderation (P25). In light of the large volume of misinformation they are required to address, "every single thing that we (fact-checkers) write has to work as hard as it possibly can" (P25). From this perspective, access to generative AI was understood as essential for sustaining fact-checking capacity under increasing verification demands.

At the same time, participants noted that access to generative AI was uneven and often constrained by factors beyond organizational control. In some countries, widely used LLM chatbots, such as ChatGPT and Gemini, were reported to be inaccessible due to IP blocking, which constrained access (P21). As a result, access to LLMs was uneven and contingent upon geopolitical and infrastructural conditions rather than organizational needs alone.

*4.4.2 Cost.* The costs associated with access and long-term adoption of LLMs continue to present a major challenge for many organizations [80], and this challenge is even greater for fact-checking organizations that already operate under financial constraints. Several participants noted that, even when tools were found to be effective, their costs placed them out of reach for many organizations. As P15 explained, although Full Fact AI was tested during the elections in their country and found to be "very effective, but the cost was a lot, (...) and most fact-checking organizations cannot afford it." The participant further noted that AI tools "are still very expensive," and that uncertainty around whether these tools will become "more expensive or cheaper" directly affects both "the speed of adoption" and "what kind of AI tools" organizations are able to adopt.

Cost pressures have increased due to funding instability. Meta's decision to end its fact-checking program in the U.S. has raised concerns about possible funding cuts in other regions [58]. This has created uncertainty for many organizations that rely on Meta as their main funder and has limited their ability to plan future investments. As P20 noted, "we'll have a financial crisis next year if Meta ends the partnership… (so) it's difficult for us at the moment to develop something new."

In response to these constraints, some organizations described strategic choices that shaped how they engaged with generative AI. As P05 mentioned, they were "very clear from day one" that they did not want to focus on "fine-tuning or trying to create a new model," because "it is very expensive," making such an investment impractical. Instead, they focused on the quality of their data. Taken together, cost considerations shape not only immediate access to tools, but also longer-term strategic decisions about whether and how generative AI adoption is viable.

*4.4.3 Organizational Capacity.* Beyond access and cost, participants emphasized organizational capacity as a key condition shaping the adoption and maintenance of generative AI systems. Participants described different modes of engagement with LLMs. Organizations with small teams and no dedicated technical staff typically rely on standard prompting. In contrast, some participants reported having in-house technical teams that include developers, product managers, and data science specialists (P02, P05, P06, P08, P09, P11, P14, P17, P25, P26). These teams play a central role in building and maintaining generative AI-based systems, including RAG pipelines and custom workflows, often supported by funding from major funders such as Google and IFCN. This variation illustrates that fact-checking organizations are not homogeneous in how they engage with generative AI: while some work primarily as end users, others actively develop, customize, and maintain in-house systems.

Some organizations relied on technical support from other organizations. Even in such cases, participants noted that generative AI-powered tools still required significant effort from their editorial teams due to rapidly changing policy contexts. As P17 explained, "we didn't expect that so many things were going to change. (...) We realized we needed to update (the chatbot) on a daily basis," a pace their team could not sustain. Instead, they have updated the system roughly every ten days in collaboration with the developers.

This burden was often unanticipated during development and placed additional strain on small teams with limited resources. Organizational capacity not only shapes whether generative AI systems can be adopted, but also whether they can be maintained over time.

## 4.5 Shifts in the Verification Ecosystem

Participants described how generative AI is reshaping the verification ecosystem through changes in public verification practices, chatbot use, and the detection of AI-generated content.

*4.5.1 Public Reliance on AI for Verification.* Participants expressed concern that individuals are increasingly turning to generative AI chatbots, such as Grok on X [97], to "fact-check" claims directly (P18, P19, P21). Despite the well-known limitations, the immediacy and fluency of generative AI chatbot responses may be particularly tempting for users [101]. As P21 observed, "I see so many people on X asking Grok, 'Is this true? Fact-check this for me.' They could easily find a credible fact-checking report with a quick search, but they don't." As a result, participants worried that LLM-infused chatbots may bypass established fact-checking practices and weaken public understanding of what fact-checking entails.

These concerns were further highlighted by recent platform developments. In June 2025, X announced the integration of AI-written notes into Community Notes, where LLMs assist in drafting notes while human contributors continue to evaluate their helpfulness [64]. Participants saw gains in speed and scale, but questioned "whether human oversight would function as intended, raising concerns that LLMs may ultimately be left to decide what counts as truth" (P19).

*4.5.2 The Evolving Role of Chatbots.* Chatbots have long been part of the fact-checking ecosystem, particularly through rule-based tip lines on WhatsApp [94], Messenger, Telegram, Line, or Viber [70]. These chatbots are used to distribute verified information and respond to audience queries. Several organizations reported that they handled between dozens and hundreds of claims per day or week, in some cases exceeding their capacity for manual response (P05, P08, P11, P12, P14, P16, P17, P22–P24). For some organizations, WhatsApp alone accounted for a substantial share of user-submitted claims, around 30–40% (P12), with user bases ranging from a few thousand (P12) to over 150,000 unique users (P26).

With the advent of LLMs, chatbots have increasingly emerged as interactive verification interfaces. Several participants mentioned the Global Fact-Check Bot project [36], which aims to enable user-friendly, cross-language information verification for fact-checkers and, subsequently, the public. Some participants reported that their organizations had already deployed public-facing generative AI-powered chatbots or were actively developing them (P05, P08, P11, P12, P14, P16, P17, P22–P24). In some cases, organizations built specialized bots, such as ChatMigrante, which provides immigration-related information in Spanish for Latino communities in the U.S [33]. All reported LLM-based chatbots rely on RAG, which allows organizations to constrain chatbot responses to curated databases of verified fact-checks and address key concerns related to hallucination. Several organizations implemented internal testing protocols whenever models or prompts were updated "to ensure it (chatbot) is within the quality parameters" (P26), alongside collecting user feedback to identify errors or limitations.

These practices illustrate how organizations seek to balance the affordances of LLM-based chatbots with ongoing concerns around accuracy and editorial responsibility. More broadly, participants noted that LLM-based chatbots reposition fact-checking as a service-oriented interaction embedded in everyday information practices. Rather than expecting audiences to read long articles, chatbots enable users to ask direct questions. Convenience was highlighted as central to this shift, with users often seeking a quick way to know whether something is true. One participant reported more positive user feedback for the LLM-based chatbot compared to the rule-based version (P26). At the same time, participants emphasized that chatbots complement rather than replace article-based fact-checking, which remains the epistemic foundation of verification work.

Despite the growing adoption of chatbots, some organizations have not invested in such tools due to limited technical capacity or different priorities. Several smaller organizations emphasized that resource limitations shaped these decisions, noting that constrained budgets were often directed toward sustaining core operations, such as paying staff salaries, rather than investing in new technological infrastructure. A small number of participants framed their decisions as strategic choices rather than capacity limitations. As one participant explained, developing an LLM-based chatbot was "not a priority at the moment," reflecting a deliberate focus on "building the tools to help with the process of what to check" (P25). This framing stood in contrast to the more widespread enthusiasm for chatbots evident elsewhere in the interviews. With resources already scarce, this view underscores the need to prioritize tools that strengthen verification itself.

*4.5.3 AI-generated Content Detection.* Participants highlighted the dual-use nature of generative AI, that it "can help fact-checkers, but it can also help people who spread misinformation," for example, by reframing claims in more "scandalous" ways to increase virality (P03). Many participants shared this concern, and framed AI-generated misinformation less as a problem of scale, in line with previous research [21], and more as a challenge of determining what is real, given the growing realism of AI-generated content, which undermines visual cues that fact-checkers had previously relied on to assess authenticity. Audio content was described as especially difficult to verify, as one participant explained: "unlike video and image, audio will not have any provenance" (P05).

Participants also expressed skepticism about the reliability of AI-generated content detection tools. Such tools typically return probabilistic outputs, such as confidence or likelihood scores [55], which were described as difficult to interpret in practice (P02, P03, P07, P14, P18, P19). This uncertainty was compounded when different tools produced conflicting results for the same piece of content. In one case, a participant published a fact-check stating that an image was AI-generated based on multiple detection tools, only to later discover that the image was authentic. Reflecting on this experience, they noted that the pressure to reach a definitive conclusion led them to treat AI detection as a binary decision rather than acknowledge conflicting results across tools: "Ultimately, I made the mistake of thinking that I had to take a stance on whether or not it's AI, rather than saying that we are getting different results from different tools" (P14).

Freely available detection tools were commonly perceived as inadequate, while more reliable systems were often accessible only through external experts or research labs, many of which were located in the Global North. Participants highlighted the difficulty of obtaining timely expert input, noting that verification work is shaped by virality and urgency, whereas responses from academic or technical experts are often not timely. The need to explain local political, cultural, or linguistic contexts to experts further delayed verification, making such reliance impractical in rapidly unfolding misinformation cycles. Participants also noted that some detection tools developed during election periods were discontinued once elections ended.

In response, one participant suggested shifting away from detectors toward content provenance approaches:

> Platform(s) should declare that the content is being created by models. So the SynthID that Google has done, I think, is really good. (...) I don't trust detectors. Some of them are very good, but they're very hard to use. They're inconsistent, and it makes it very difficult for us to explain to our users how we came to a conclusion. Fact-checks have to be transparent. If we say, 'Oh, we just trusted this thing, and it gave us a 67% score,' that's not helpful. Everything we do has to be reproducible. (P25)

Rather than trying to infer whether content is AI-generated using probabilistic results, this participant argued that AI companies should disclose when content is created by AI systems, viewing disclosure as a more transparent and reliable alternative to detection.

## 5 Discussion

Here we reflect on how fact-checkers use generative AI in practice and the benefits and challenges they encounter **(RQ1)**, how they perceive its impact on verification work **(RQ2)**, and the conditions shaping adoption **(RQ3)**. Our findings suggest that LLM adoption in fact-checking is not determined primarily by technical capability, but by an ongoing negotiation between epistemic risk, professional norms and structural constraints. Across interviews, participants rejected end-to-end automation and instead favored modular, human-in-the-loop tools that integrate into existing workflows, aligning with previous research emphasizing augmentation over automation [73, 91, 95]. At the same time, rather than simply confirming a shift from automation to augmentation, our findings show that augmentation is tightly bounded. LLMs are incorporated into peripheral and preparatory stages of the workflow, such as monitoring, transcription, summarization, initial research, writing support, and content production for distribution. More autonomous creative companion or processing agent configurations are reserved for low-stakes or side tasks [65]. This situates the use of LLMs at the lower-autonomy end of the human–AI collaboration spectrum [60]. Epistemically decisive tasks, such as evaluating evidence and issuing verdicts, remain explicitly protected from automation. This positions LLMs not as epistemic agents, but as infrastructural supports, whose outputs remain subordinate to human judgment in contexts where accountability cannot be delegated.

Participants' accounts reveal both the opportunities that generative AI offers and the challenges it introduces across verification work. On the benefits side, generative AI was associated with time savings and increased productivity through assistive tasks, as well as expanding verification capacity and enabling new interaction modes, such as public-facing chatbots. These findings provide empirical evidence for what prior work has suggested in principle: that LLMs can assist fact-checkers across various stages of verification, including claim detection, evidence retrieval, and explanation generation [8, 23, 25]. Beyond these end user affordances, our findings show that some organizations have also begun using LLMs to develop and maintain their own in-house tools and infrastructures. On the challenges side, factuality, especially hallucinations, was described as the most serious concern surrounding the use of LLMs in fact-checking. Participants framed the tolerability of inaccuracies in LLM outputs as context-dependent, varying by audience and use case. In professional settings, a lower accuracy threshold is workable because fact-checkers, as expert users, investigate claims and catch errors before they reach final verdicts. Public-facing applications such as chatbots, by contrast, demand far higher accuracy thresholds, since general audiences typically lack the verification practices needed to recognize inaccurate outputs. This risk is compounded by the fluent and confident style of LLM-generated text [8], which can lead non-expert users to accept false or misleading information as credible.

Our interviews also identify conflicts between generative AI and fact-checking principles, pointing to a mismatch between how LLMs work and what fact-checking requires, particularly around fairness, accountability, and transparency. Fairness concerns are most pronounced in low-resource languages. In these contexts, transcription and evidence retrieval often perform poorly and, in some cases, introduce new misinformation into workflows. These accounts point an underlying bias in the models themselves [10], shaping both which languages are supported and what evidence surfaces. AI-assisted fact-checking has already been shown to benefit majority communities more than minority ones, leaving misinformation that targets minority groups to circulate unchecked [76]. Our findings suggest a similar asymmetry at the language level, where the language of some communities is simply harder to verify. This indicates that verification capacity is unevenly distributed across linguistic contexts, making some claims more difficult to reliably verify than others. When tools work better in some languages than others, fact-checking itself reaches some communities more reliably than others. A further difficulty arises around accountability. Fact-checkers are fully accountable for the accuracy and implications of their outputs, a responsibility that risks being eroded when delegated to black box systems. Participants therefore rejected the idea of treating AI systems as independent decision-makers, preferring to maintain humans in the loop and attach accountability to practitioners rather than systems. Closely linked to accountability, transparency was described as harder to maintain. How the model explains itself often differs from what actually drives its output [100], which conflicts with fact-checkers' need to show how they reached their verdict [52]. Another transparency challenge arises at the level of disclosure. As generative AI becomes embedded in everyday tools, it is increasingly unclear what should count as "AI use" and it is therefore harder to be transparent about it.

Perceived impacts extend beyond internal workflows to broader ecosystem shifts. Audiences increasingly turn to general-purpose LLM chatbots, such as Grok on X, to fact-check claims directly [81], which participants worried may erode public understanding of what fact-checking entails. New features such as AI-written Community Notes [64] raise further questions about whether human oversight will retain its role as LLMs take on a more active role in evaluating social media posts. Another shift concerns fact-checking organizations themselves, which are developing LLM-based chatbots that mainly support the distribution of previously fact-checked claims.

Participants were divided on whether this represents the right investment. Chatbots expand access to existing fact-checks, but the main bottleneck lies in the verification pipeline itself; in that sense, investment in core verification work aligns more closely with where the field's capacity constraints actually lie. A further shift concerns AI-generated content detection, where tools return probabilistic outputs that are difficult to interpret and sometimes differ across tools. These outputs clash with the way that fact-checking procedures arrive at a final verdict, which requires clear, reproducible reasoning that can be explained to audiences. Together, these dynamics suggest that generative AI is not only transforming how fact-checkers work, but also reshaping the wider ecosystem in which verification takes place. Verification functions are increasingly distributed across actors who may not share fact-checking's professional norms, raising questions about which standards should apply, who should set them, and how established fact-checking principles can be sustained as the boundaries of verification expand.

Across the workflow integration, conflicts, and ecosystem shifts described above, a broader pattern emerges: human–AI interaction in fact-checking is highly heterogeneous and structurally shaped by the conditions under which organizations operate. This variation is produced by three interlocking conditions. Access is uneven and sometimes contingent on geopolitical factors outside organizational control. Cost influences both immediate adoption and long-term sustainability, particularly amid funding instability following Meta's decision to end its U.S. third-party fact-checking program [58] and the uncertainty this creates elsewhere. Organizational capacity determines not only whether generative AI systems can be adopted, but also whether they can be sustained over time. Together, these conditions shape which organizations can benefit from generative AI, for how long, and on what terms.

These findings extend prior research in two ways. First, prior work has largely treated fact-checkers as the end users of AI systems. Our findings show that this distinction is no longer sufficient. Some organizations now have in-house development teams building and maintaining their own LLM-based infrastructures. For instance, advanced techniques such as RAG, framed as inaccessible to fact-checkers [30], or something that was envisioned [95], are being operationalized in everyday verification workflows. Second, while existing research assumes that improved alignment with workflows will increase adoption, our findings demonstrate that adoption is also constrained by professional norms, epistemic risk, and structural conditions. Even high-performing tools may be rejected or restricted when they conflict with professional standards or exceed organizational capacity. Taken together, human–AI interaction in fact-checking is shaped by technological affordances, professional norms, access limitations, costs, organizational capacity, and the broader ecosystem shifts. This variation has important implications, as it affects who can experiment with, govern, and benefit from AI-assisted verification. Rather than a uniform trajectory toward automation, the integration of generative AI in fact-checking is fragmented, negotiated, and uneven. Designing for this space therefore requires not only improving model performance, but aligning systems with professional norms, supporting diverse organizational capacities, and accounting for the broader redistribution of verification across the information ecosystem.

## 5.1 User-centered Design Implications for Fact-checking

Based on our findings, we identify the following design implications: (1) support narrative-level analysis to identify recurring themes across platforms; (2) enable cross-organizational verification infrastructure for collaboration and reuse; (3) allow multimodal submissions in verification tip lines; (4) develop transcription tools that handle accent variation, code-switching, and low-resource languages; and (5) adapt to changing information consumption by building on existing rule-based systems for rapid information dissemination. Appendix C summarizes these implications.

### 5.2 Limitations and Future Work

These findings should be interpreted in light of a few limitations. First, interviews were conducted in English, which may have discouraged participation or influenced how comfortably non-native speakers expressed themselves. Second, participants were selected to include fact-checkers who use generative AI in their daily work. While this provided experience-based insights into both perceived benefits and challenges, it may underrepresent perspectives from practitioners with very little or no direct exposure to these tools. Third, focusing on participants from IFCN signatory organizations may exclude practices present in the wider fact-checking community, approximately 65% of which are not IFCN signatories [31]. As a result, the findings may not fully reflect the diversity of fact-checking practices across different organizational contexts. Future research could address these limitations by extending interviews beyond English and including participants with limited or no experience with generative AI, as well as those from organizations that are not IFCN signatories.

## 6 Conclusion

Drawing on interviews conducted across diverse organizational and regional contexts, we demonstrate that LLM adoption in fact-checking is highly heterogeneous, ranging from standard prompting practices to in-house system-level approaches. Participants rejected the idea of end-to-end automation, instead emphasizing modular human-in-the-loop uses that preserve professional judgment and align with commitments to transparency, accountability, and fairness. By highlighting differences in organizational capacity and structural inequalities, this work challenges homogeneous and purely technical narratives of AI adoption and highlights the uneven ways in which generative AI is reshaping verification work. These findings suggest that tools designed to support verification should account for organizational contexts, existing workflows, and professional norms, rather than assuming a one-size-fits-all approach.

## Generative AI Usage Statement

In accordance with the ACM policies [1, 68], the first author used ChatGPT-5.1 to assist with the grammar and fluency of the writing. The second and third authors did not use any generative AI tools.

## Ethical Considerations Statement

The study received ethical approval from the Oxford Internet Institute Departmental Research Ethics Committee, University of Oxford (Reference number: 706386). Informed oral consent was obtained from all participants before data collection. Participants' gender was self-reported.

This study may have unintended impacts if its findings are interpreted as advocating uncritical or uniform reliance on LLMs for fact-checking across organizations. To mitigate this risk, we emphasize heterogeneity, structural constraints, and professional norms throughout the paper, and avoid one-size-fits-all recommendations. By highlighting the conflicts between fact-checking principles and generative AI, we aim to discourage technosolutionist interpretations and support solutions guided by commitments to fairness, accountability, and transparency.

## Acknowledgments

We thank the participants for their valuable time and contributions to this study.
This research was supported by the Oxford Internet Institute DPhil Student Research Fund at the University of Oxford, which covered participant compensation for interviews.

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# A Study Materials

## A.1 Participant Demographics

Table 1 presents an overview of participant demographics. Positions were aggregated into broader functional role categories in order to reduce the risk of re-identification, particularly for senior or unique roles. Regions reflect the area(s) in which participants' fact-checking organizations operate, rather than individual countries, to preserve participant anonymity.

## A.2 Recruitment Strategy

Our recruitment strategy combined purposive sampling, snowball sampling, and listserv recruitment strategies [77]. As this study focused on AI in fact-checking, we purposively identified organizations with relevant experience. To do so, we reviewed grant funding records (e.g., Global Fact Check Fund [50] and JournalismAI Innovation Challenge [56]), consulted the programs of fact-checking conferences such as Global Fact, as well as previous research and reports on AI in fact-checking [34]. Identified fact-checkers were contacted directly via email whenever possible, and via LinkedIn when direct contact information was unavailable. In addition, at the end of each interview, participants were invited to suggest further potential interviewees, allowing us to expand the sample through snowball sampling. As noted in previous research, recruiting fact-checkers was also challenging in our research [73, 91]. To expand our reach, we also relied on professional networks [32, 45, 78] and circulated study advertisements via their mailing lists. Recruitment materials included a brief description of the study and an online registration form, provided below (Appendix A.3).

Following Braun and Clarke [14], the number of participants in our study (N=29) was guided by the concept of information power. Information power suggests that the more relevant information a sample carries, the fewer participants are needed [66]. We recruited participants with experience of generative AI use in fact-checking,

Table 1. Participant Demographics

| ID | Gender | Age group | Highest education | Fact-checking experience | Role category | Region |
|---|---|---|---|---|---|---|
| P01 | Woman | 25–34 | Master's degree | 4–6 years | Editorial | Western Asia |
| P02 | Man | >55 | Master's degree | >10 years | Editorial | Western Europe |
| P03 | Woman | 25–34 | Bachelor's degree | <1 year | Editorial | Eastern Europe |
| P04 | Woman | 25–34 | Bachelor's degree | 4–6 years | Editorial | Western Asia |
| P05 | Man | 35–44 | Bachelor's degree | 7–10 years | Editorial | Southern Asia |
| P06 | Man | 25–34 | Bachelor's degree | 7–10 years | Technical | Southern Europe |
| P07 | Woman | 45–54 | Master's degree | 4–6 years | Editorial | Northern Europe |
| P08 | Woman | 45–54 | Bachelor's degree | <1 year | Technical | South America |
| P09 | Man | 18–24 | Bachelor's degree | 1–3 years | Technical | Western Africa; Middle Africa |
| P10 | Woman | 25–34 | Bachelor's degree | 1–3 years | Editorial | Western Asia |
| P11 | Woman | 25–34 | Bachelor's degree | 7–10 years | Editorial | South-Eastern Asia |
| P12 | Man | 25–34 | Master's degree | 7–10 years | Editorial | South America |
| P13 | Woman | 35–44 | Master's degree | 7–10 years | Editorial | Eastern Europe |
| P14 | Woman | 35–44 | Postgraduate diploma | 7–10 years | Editorial | Southern Asia; South-eastern Asia |
| P15 | Man | 25–34 | Master's degree | 7–10 years | Editorial | Western Africa |
| P16 | Woman | 35–44 | Master's degree | 1–3 years | Editorial | South America |
| P17 | Woman | 45–54 | Bachelor's degree | 4–6 years | Editorial | Northern America |
| P18 | Woman | 25–34 | Master's degree | 4–6 years | Editorial | Western Africa; Eastern Africa; Southern Africa |
| P19 | Woman | 25–34 | Master's degree | 4–6 years | Editorial | Western Africa; Eastern Africa; Southern Africa |
| P20 | Woman | 35–44 | Bachelor's degree | 4–6 years | Editorial | Eastern Asia |
| P21 | Woman | 25–34 | Master's degree | 4–6 years | Editorial | Eastern Asia |
| P22 | Man | 25–34 | Master's degree | 1–3 years | Editorial | Central Asia; Eastern Asia |
| P23 | Woman | 35–44 | Master's degree | 1–3 years | Editorial | South-Eastern Asia |
| P24 | Man | 25–34 | Master's degree | 4–6 years | Editorial | Southern Asia |
| P25 | Man | 45–54 | Bachelor's degree | 4–6 years | Technical | Northern Europe |
| P26 | Man | 35–44 | Master's degree | 4–6 years | Technical | South America |
| P27 | Woman | 35–44 | Master's degree | 1–3 years | Editorial | Southern Asia |
| P28 | Man | 35–44 | Doctorate degree | 1–3 years | Editorial | Southern Europe |
| P29 | Man | >55 | Master's degree | >10 years | Editorial | Northern America |

across a range of organizational and geographic contexts, and stopped recruitment when further interviews began to confirm rather than extend the patterns we were seeing in earlier ones. Our study also includes more participants than the most common sample size of 12 reported for HCI studies [20].

### A.3 Participant Sign-Up Survey for the Study

(1) Are you a fact-checker? (Yes / No)
(2) Full name
(3) Email address
(4) Name of the organization you work for
(5) Have you used artificial intelligence (AI) tools, including generative AI tools (e.g., ChatGPT, Bard, Claude, Microsoft Copilot), in your fact-checking work? (Yes / No)
(6) If yes, which AI tools have you used?
(7) Link to professional profile (e.g., fact-checking website, LinkedIn)
(8) How did you hear about this study? (IFCN email list; EFCSN email list; Hacks/Hackers email list; contacted directly by the researcher; other)

### A.4 Demographic Questionnaire

(1) Full name
(2) Email address
(3) Gender (Woman; Man; Non-binary; Prefer not to say; Other)
(4) Age (18–24; 25–34; 35–44; 45–54; > 55)
(5) Highest level of education completed (High school; Bachelor's; Master's; Doctorate; Other)
(6) Organizations currently and previously worked with as a fact-checker
(7) Length of experience as a fact-checker
(8) Specific role in current organization
(9) Length of time in current organization
(10) Main topics or domains covered by the organization (e.g., politics, health, climate, technology)
(11) Country/countries of operation
(12) Language(s) of publication

### A.5 Interview Guide

Background & Process

(1) How did you first get involved in fact-checking?
(2) Can you describe your role in the organization and how your team is structured?
(3) Which topics or areas do you most often focus on in practice? Where do you see the highest volume of misinformation?
(4) Can you walk me through your typical fact-checking process, from identifying a claim to publishing the final output?
(5) What tools or resources do you use at different stages of the process?

Chatbots & Generative AI Use

(6) Does your organization use a WhatsApp tip line or any chatbots? If yes, could you describe the main features and how they're built (AI vs. rules-based)?
(7) What advantages and challenges have you experienced with these tools?
(8) Are there any statistics or insights you can share on how they're used?

Generative AI in Fact-Checking

(9) What has your overall experience been with generative AI so far? Advantages and disadvantages?
(10) How has generative AI affected your workload, and have you noticed any differences when working in different languages?

(11) Are there any editorial or ethical guidelines you follow when using generative AI?
(12) In your view, how much of the fact-checking process could be automated, and what parts should stay human-led?

AI-generated Misinformation & Platforms

(13) How are you responding to AI-generated misinformation like deepfakes or synthetic text?
(14) What's your view on community-led models like Community Notes?
(15) It looks like LLMs will soon be integrated into Community Notes on X. What do you think about using generative AI in this context?
(16) What do you think about people using Grok on X to fact-check?

Future Directions

(17) Are there new technologies your organization is exploring for fact-checking?
(18) If you could design your ideal generative AI tool for fact-checking, what would it look like?
(19) What impacts do you forsee as generative AI becomes more embedded in fact-checking?
(20) Is there anything we haven't covered that you'd like to add?

## B Themes

Table 2. Overview of themes

| Theme | Description |
|---|---|
| Human-led fact-checking with LLM assistance | LLMs are selectively used across stages of the fact-checking workflow, including claim detection, evidence gathering, and distribution, while verification remains human-led |
| Fact-checking principles in conflict with generative AI | How principles such as fairness, accountability, and transparency shape the use of generative AI in fact-checking |
| Factuality challenges in LLM-assisted fact-checking | How hallucinations, probabilistic outputs, and knowledge limitations challenge factuality and require human oversight |
| Conditions of adoption: Access, cost, and organizational capacity | How infrastructural factors, including access, financial constraints, and organizational capacity, shape whether and how LLMs are adopted |
| Shifts in the verification ecosystem | How LLMs are reshaping public verification practices, chatbot use, and the detection of AI-generated content |

We identified five themes: Human-led fact-checking with LLM assistance and factuality challenges in LLM-assisted fact-checking address how fact-checkers use generative AI in their workflows and the benefits and challenges they encounter (RQ1). Shifts in the verification ecosystem and fact-checking principles in conflict with generative AI capture how fact-checkers perceive the impact of generative AI on verification practices (RQ2). Conditions of adoption: Access, cost, and organizational capacity highlights how organizational, technical, and editorial practices shape the adoption and use of generative AI in fact-checking contexts (RQ3). Table 2 provides an overview of all themes.

## C User-centered Design Insights

Table 3 summarizes user-centered design considerations.

Table 3. User-centered design insights grounded in fact-checkers' experiences with generative AI

| Challenges | Design implications | Implications for fact-checkers |
|---|---|---|
| Misinformation often appears as recurring narratives and travels across platforms. | Design tools for narrative-level analysis that cluster related claims and trace recurring narratives (P03, P04). | Helps fact-checkers understand how misinformation spreads and persists over time, enabling more strategic interventions. |
| Verification work is fragmented across organization-specific databases. | Design shared, cross-organizational verification infrastructures that aggregate fact-checks from multiple IFCN signatories (P13). | Reduces duplicated effort, improves consistency across fact-checks, and supports collaboration within the fact-checking ecosystem. |
| Fact-checking requests received via tip lines often involve multimodal content. | Design compatible tools that support the analysis of text, images, audio, and video (P09). | Supports timely verification in messaging environments where misinformation is rarely text-only. |
| Transcription tools often perform poorly across different English accents, low-resource languages, and code-switched speech. | Develop inclusive transcription tools that support diverse English accents, low-resource languages, and code-switched speech. (P09, P14, P15). | Enables verification work in diverse linguistic and regional contexts. |
| Information consumption is shifting away from long-form articles toward faster, conversational forms of access. | Integrate conversational chatbots into existing tip lines to provide faster, user-friendly access to verified information (P19, P27). | Reduces the effort required for fact-checkers to disseminate verified information across formats and platforms. |